Name
Date
Course

Rational Functions and Their Graphs

Rational Function: A function of the form $\frac{A(x)}{B(x)}$, where $A(x)$ and $B(x)$ are polynomials.

Examples: $f(x)=\frac{x}{x^3+6}$, $g(x)=\frac{2x^3+4x^2-2x-1}{x^2-4}$, $h(x)=\frac{-2}{x-1}$

The parent reciprocal function is the most basic type of rational function.

$$f(x)=\frac{1}{x}$$

When graphing rational functions, it is necessary to convert to "proper" form.

Proper Form of a Rational Function: When the degree of the polynomial in the numerator is less than the degree of the polynomial in the denominator.

Examples

Proper: $\frac{x-5}{x^2-2}$, $\frac{1}{x^4}$, $\frac{7x^5-2x+1}{3x^6-4x-2}$

Improper: $\frac{x^2-2}{x+6}$, $\frac{10x^4-2x^3+1}{x^3+3x^2-x+1}$, $\frac{-8x^7}{x^5+4x}$

Steps for Graphing Rational Functions

I) Find x and y intercepts and draw them on the graph.

II) Find asymptotes and draw them as dotted lines.
* The function must be in proper form.

III) Find the coordinates of holes and draw them on the graph.

IV) Create a sign chart and use it, along with the previous information, to draw the graph of the function.

The Asymptotes of a Rational Function Theorem

i) Let $f(x)=\frac{A(x)}{B(x)}$ be a rational function in proper form. If $B(c)=0$, then $x=c$ is a vertical asymptote of the graph of the function.

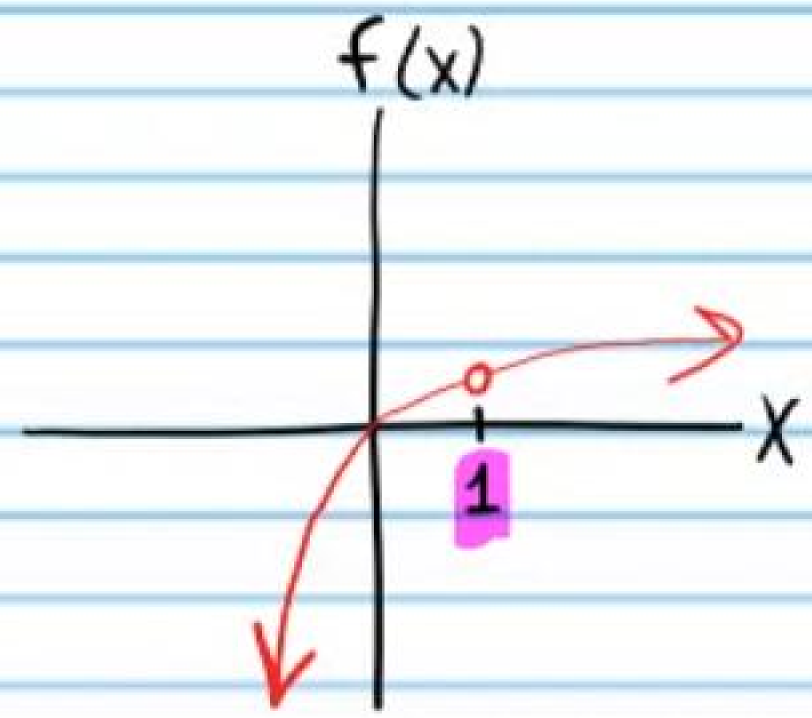
ii) If a rational function is in proper form, then there is a horizontal asymptote on the x-axis.

iii) If a rational function is in improper form, then there could be a horizontal asymptote or an oblique asymptote.

Graphical Hole: An infinitesimally small break in the continuity of a graph (drawn as a little open circle).

If $f(x)=\frac{A(x)}{B(x)}$ is a rational function, and $g(x)$ is a factor of both $A(x)$ and $B(x)$, and $g(c)=0$, then a hole (i.e. a little open circle) is drawn at the point with $x=c$.

Example: $f(x)=\frac{x^2-x}{x^2+x-2}=\frac{x(x-1)}{(x+2)(x-1)}$



Graph the following functions. Draw asymptotes as dotted lines and write their equations.

① $f(x) = \dfrac{1}{x^2-2x-3}$

$f(x) = \dfrac{1}{(x+1)(x-3)}$

Vertical Asymptote Equations

$x=-1, \quad x=3$

Horizontal Asymptote Equations

$y=0$

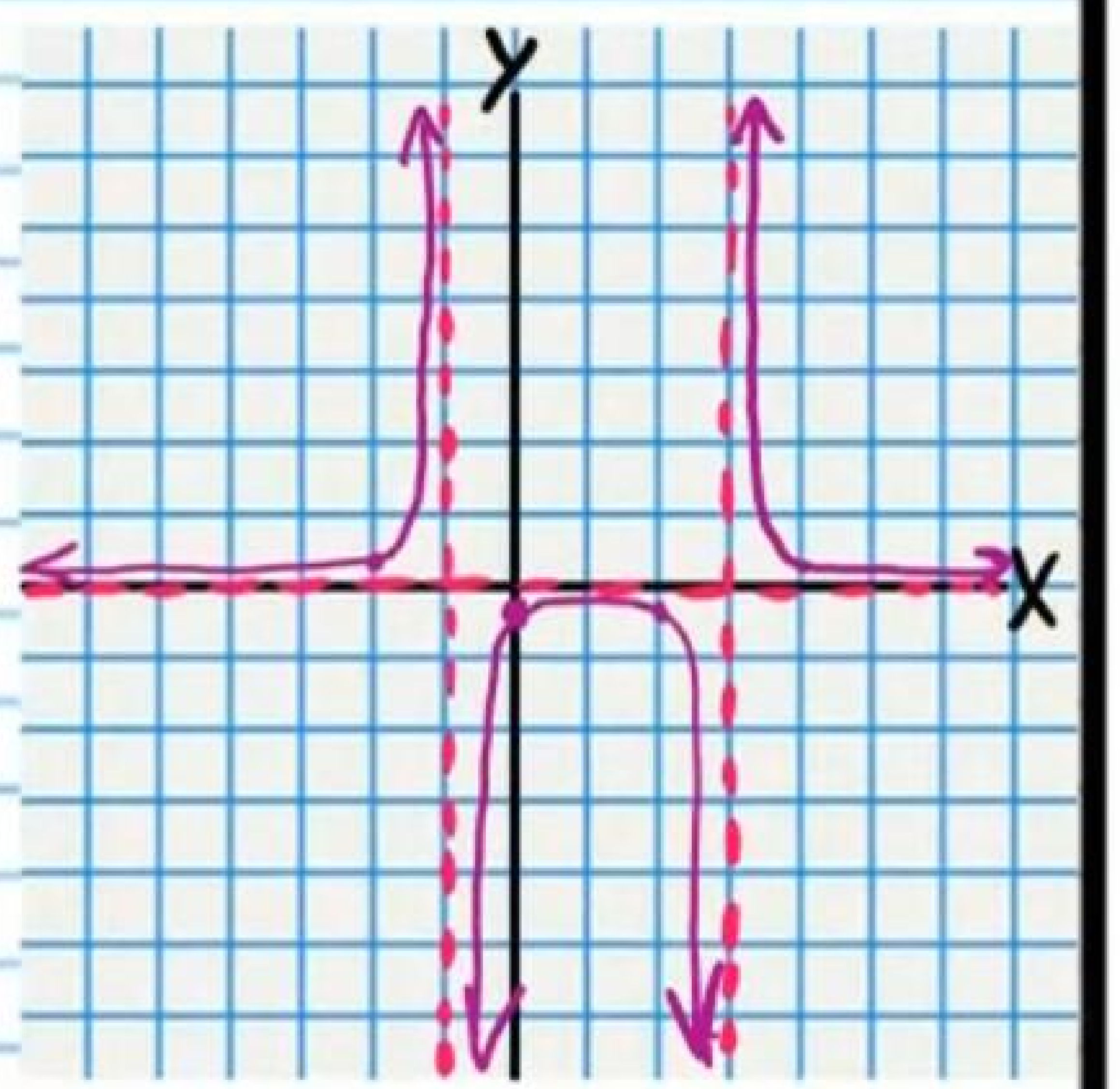
Sign Chart

+	−1	−	3	+

Y-Intercept

$f(0) = \dfrac{1}{0^2-2(0)-3} = -\dfrac{1}{3}$

$(0, -\frac{1}{3})$



X-Intercept

$f(x)=0$

$\dfrac{1}{x^2-2x-3} = 0$

Impossible.

No solution.

② $f(x) = \dfrac{-16}{x^2+4x+4}$

$= \dfrac{-16}{(x+2)(x+2)}$

Vertical Asymptote Equations

$x=-2$

Horizontal Asymptote Equations

$y=0$

Sign Chart

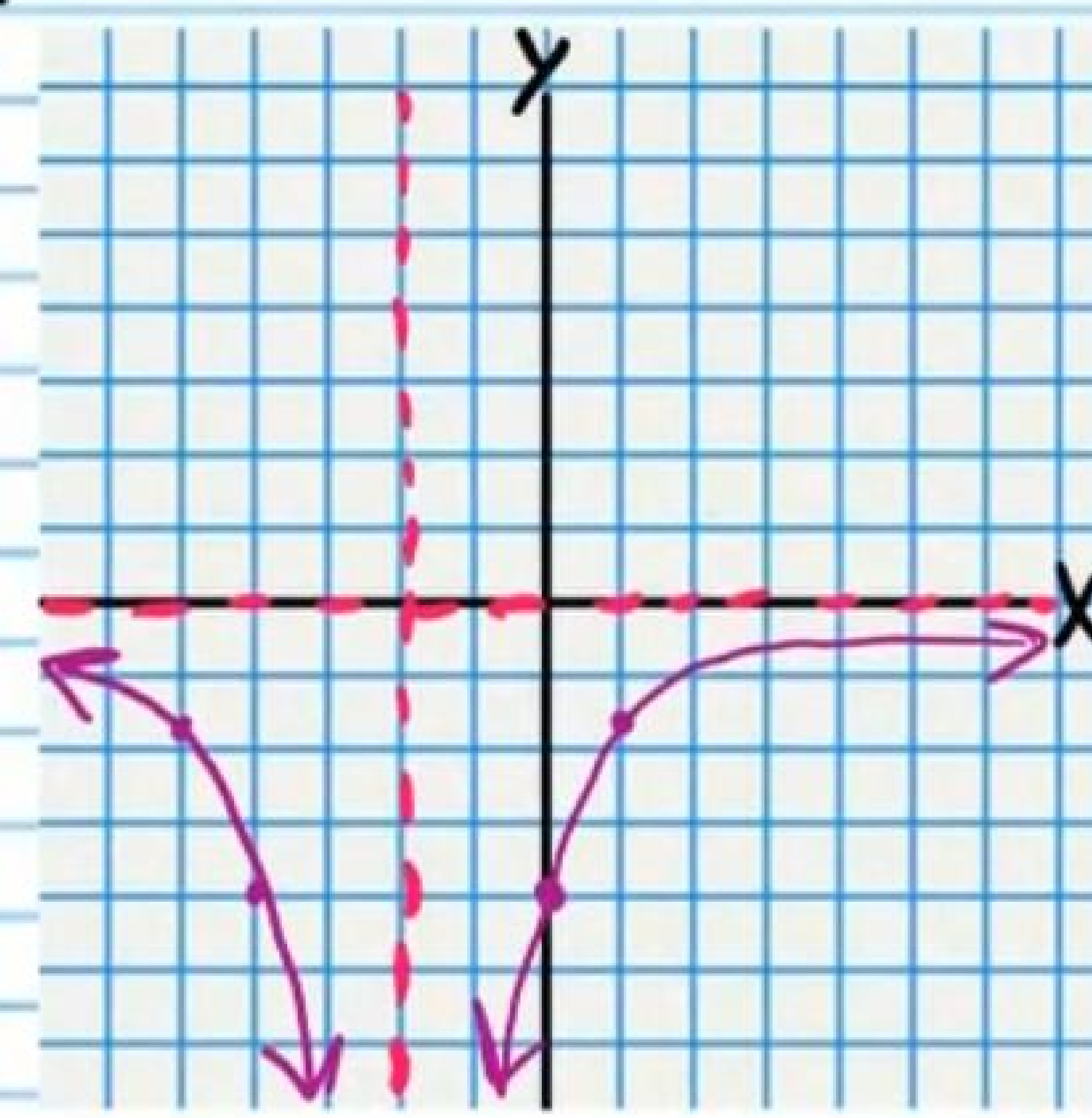
−	−2	−

Y-Intercept

$f(0) = \dfrac{-16}{0^2+4(0)+4} = \dfrac{-16}{4}$

$= -4$

$(0,-4)$



X-Intercept

None

③ $f(x) = \frac{18}{x^2+x-6}$

$= \frac{18}{(x+3)(x-2)}$

Vertical Asymptote Equations

$x = -3,\ x = 2$

Horizontal Asymptote Equations

$y = 0$

Sign Chart

$+ \quad - \quad +$

$-3 \quad 2$

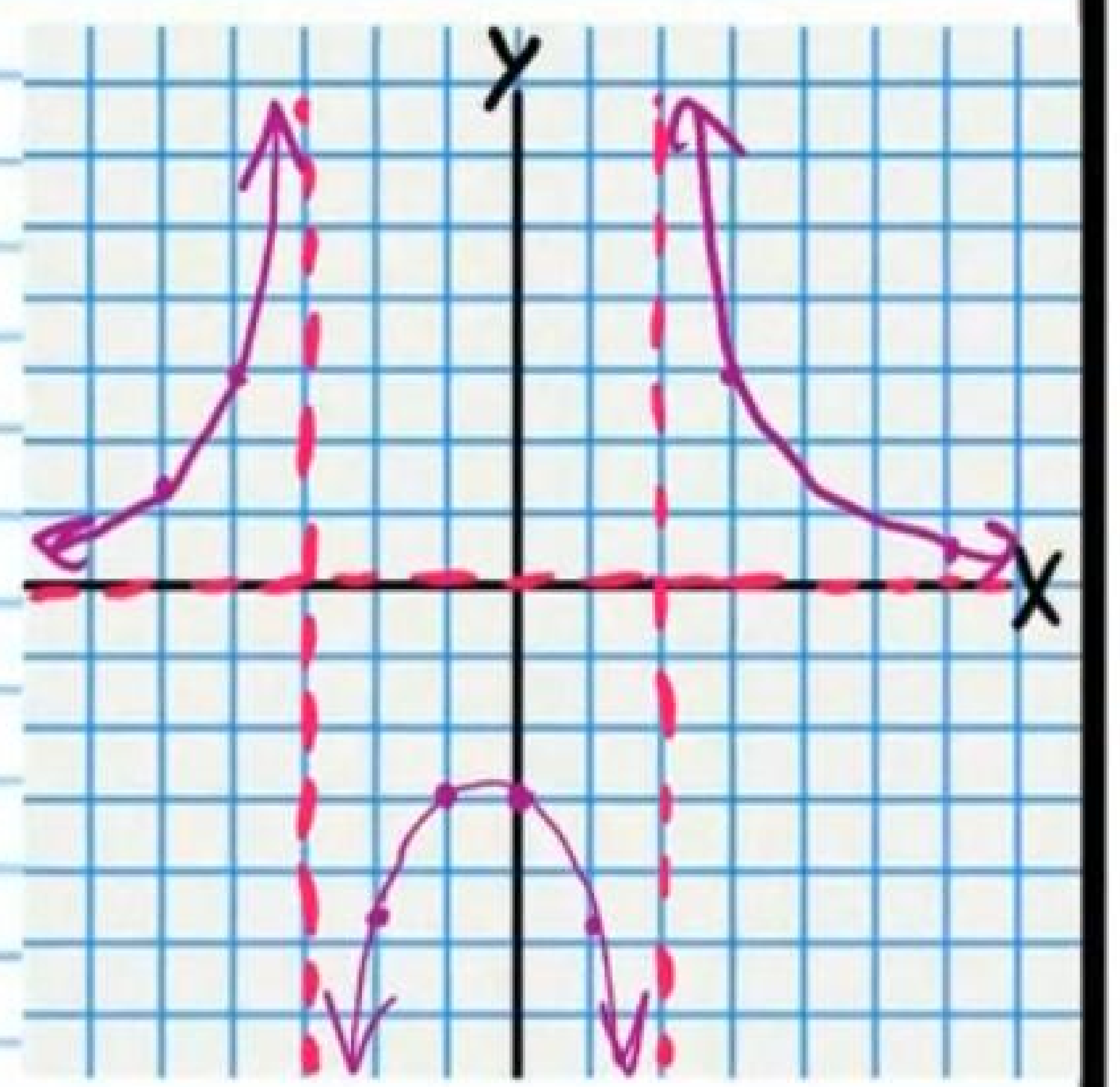
Y-Intercept

$f(0) = \frac{18}{0^2+0-6} = -3$

$(0, -3)$

X-Intercept

None.



④ $f(x) = -\frac{1}{x^2-2x+1}$

$= -\frac{1}{(x-1)(x-1)}$

Vertical Asymptote Equations

$x = 1$

Horizontal Asymptote Equations

$y = 0$

Sign Chart

$- \quad -$

1

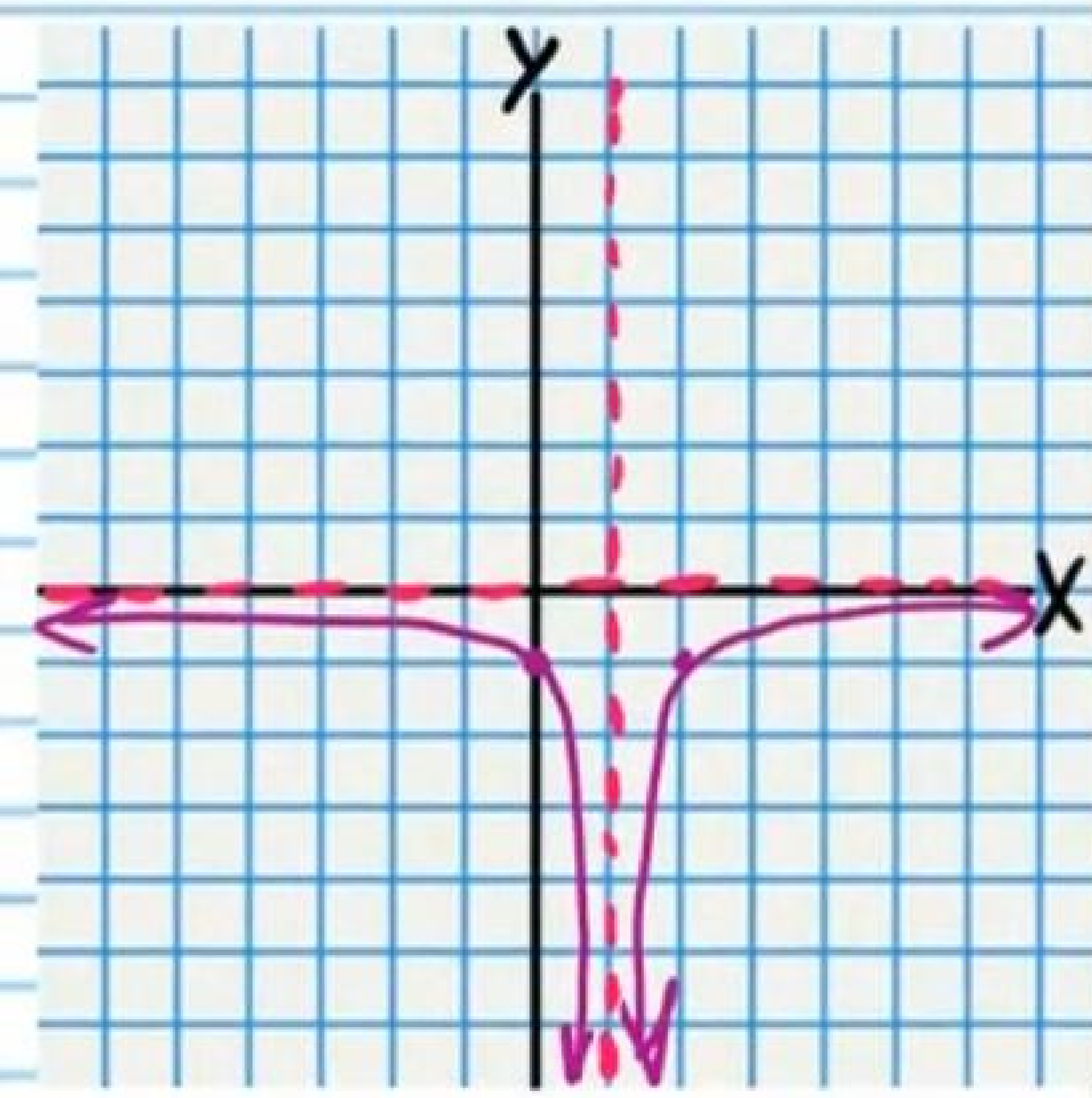
Y-Intercept

$f(0) = -\frac{1}{0^2-2(0)+1} = -1$

$(0, -1)$

X-Intercept

None



⑤ $f(x) = \frac{x+2}{x^2+2x-15}$

$= \frac{x+2}{(x+5)(x-3)}$

Vertical Asymptote Equations

$x=-5, \; x=3$

Horizontal Asymptote Equations

$y=0$

Sign Chart

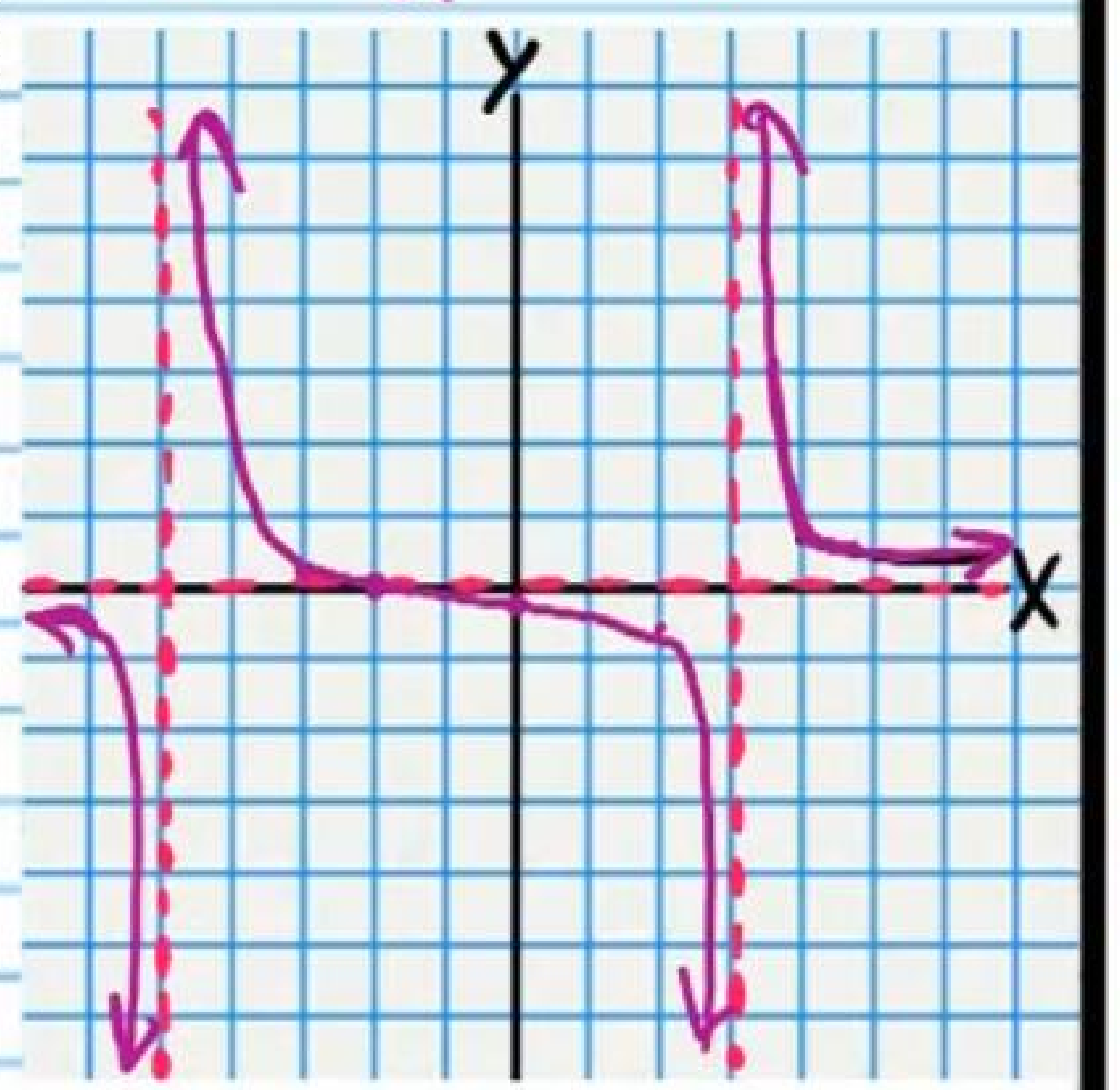
$-$ | $+$ | $-$ | $+$

-5 $\quad -2 \quad$ 3

Y-Intercept

$f(0) = \frac{0+2}{0^2+2(0)-15} = -\frac{2}{15}$

$(0, -\frac{2}{15})$



X-Intercept

$f(x)=0$

$\frac{x+2}{x^2+2x-15} = 0$

$x+2=0$

$x=-2$

$(-2,0)$

⑥ $f(x) = \frac{x-3}{x^2+2x+1}$

$= \frac{x-3}{(x+1)(x+1)}$

Vertical Asymptote Equations

$x=-1$

Horizontal Asymptote Equations

$y=0$

Sign Chart

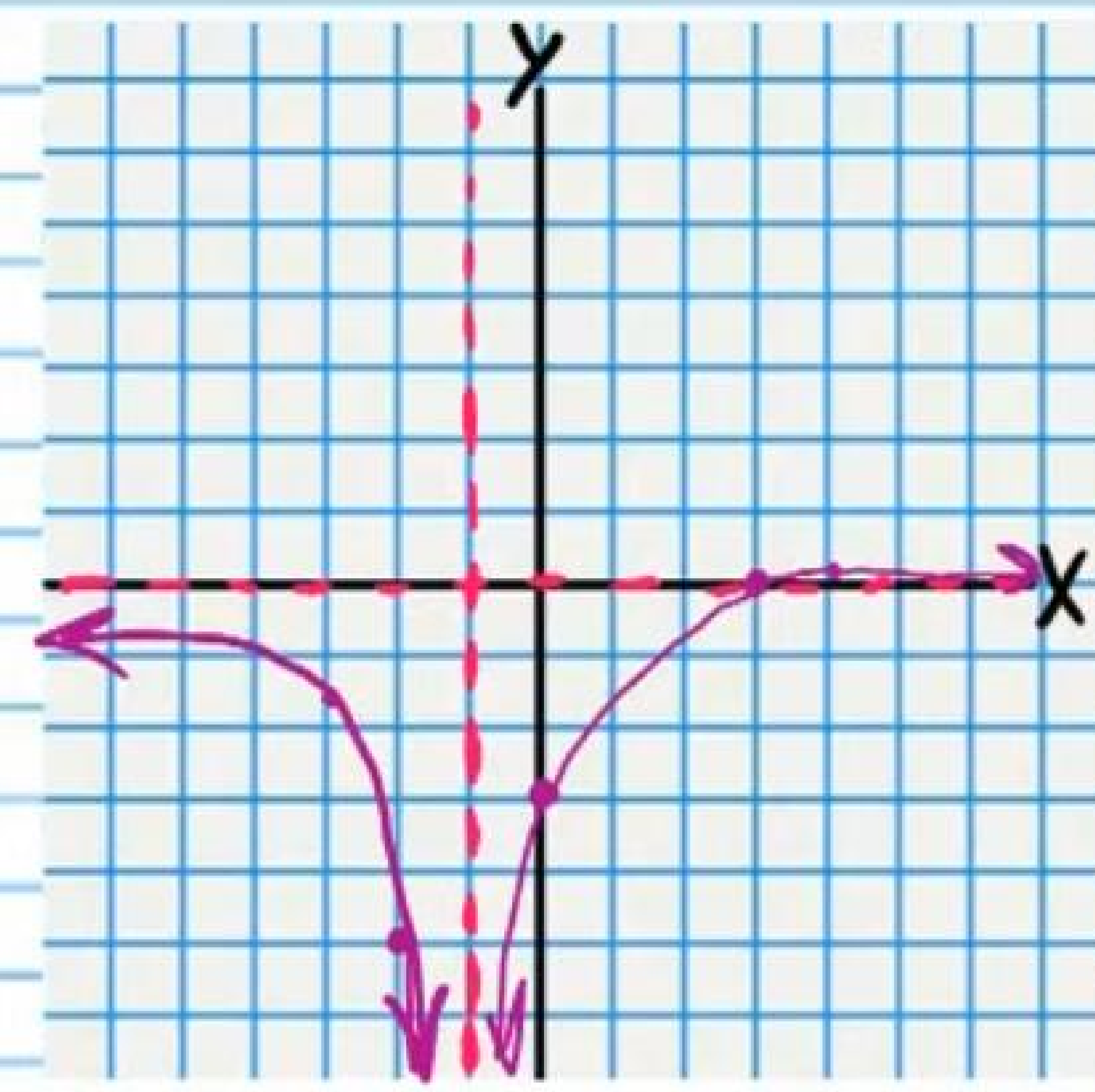
$-$ | $-$ | $+$

$-1 \quad 3$

Y-Intercept

$f(0) = \frac{0-3}{0^2+2(0)+1} = -3$

$(0,-3)$



X-Intercept

$f(x)=0$

$\frac{x-3}{x^2+2x+1} = 0$

$x-3=0$

$x=3 \quad (3,0)$

⑦ $f(x) = \frac{x-4}{x^2-5x-6}$

$= \frac{x-4}{(x-6)(x+1)}$

Vertical Asymptote Equations

$x=6,\ x=-1$

Horizontal Asymptote Equations

$y=0$

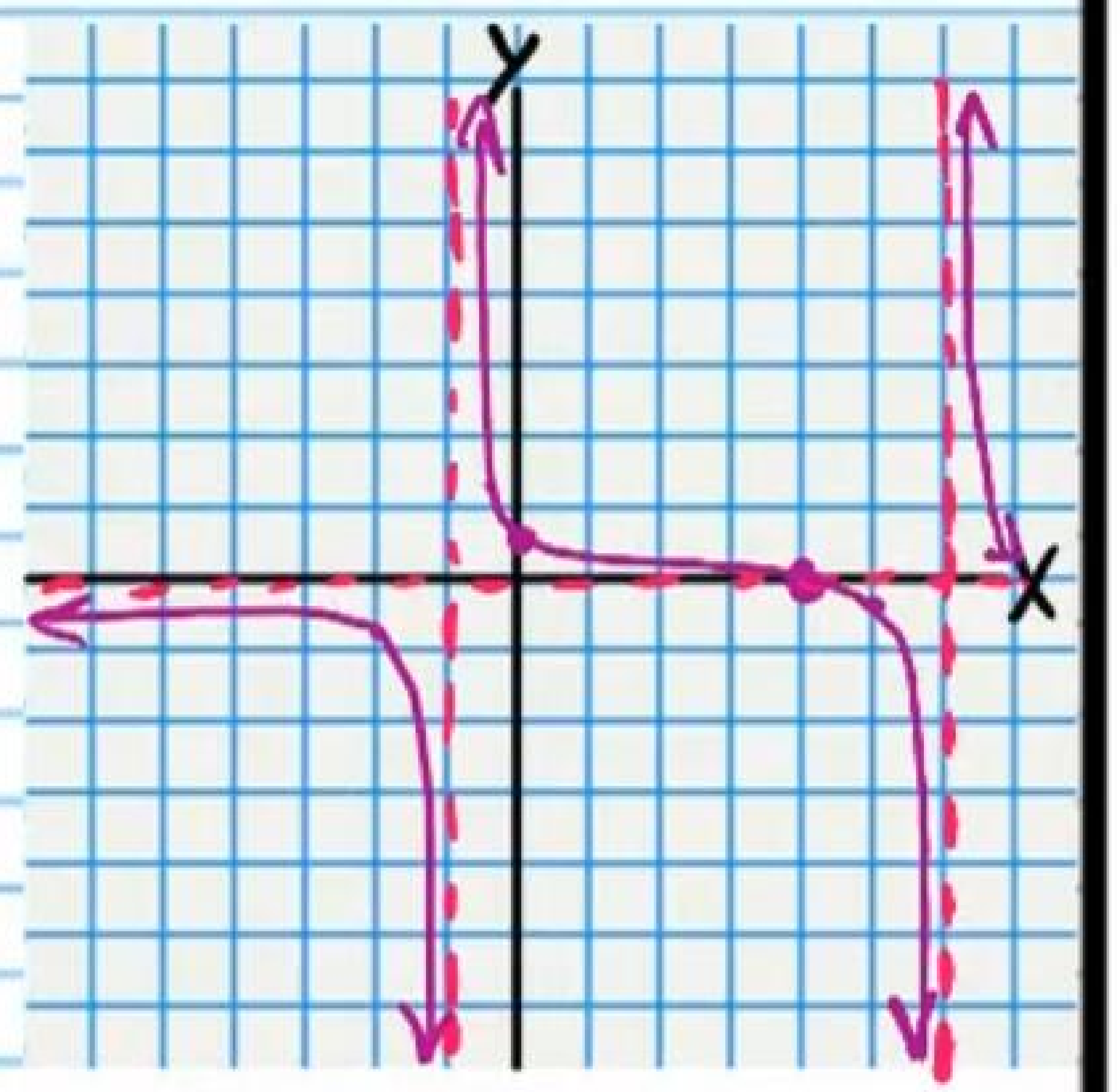
Sign Chart

$-$ | -1 | $+$ | 4 | $-$ | 6 | $+$

Y-Intercept

$f(0) = \frac{0-4}{0^2-5(0)-6} = \frac{2}{3}$

$\left(0, \frac{2}{3}\right)$



X-Intercept

$f(x)=0$

$\frac{x-4}{x^2-5x-6} = 0$

$x-4=0$

$x=4 \quad (4,0)$

⑧ $f(x) = -\frac{2x+6}{x^2+5x+4}$

$= -\frac{2x+6}{(x+4)(x+1)}$

Vertical Asymptote Equations

$x=-4,\ x=-1$

Horizontal Asymptote Equations

$y=0$

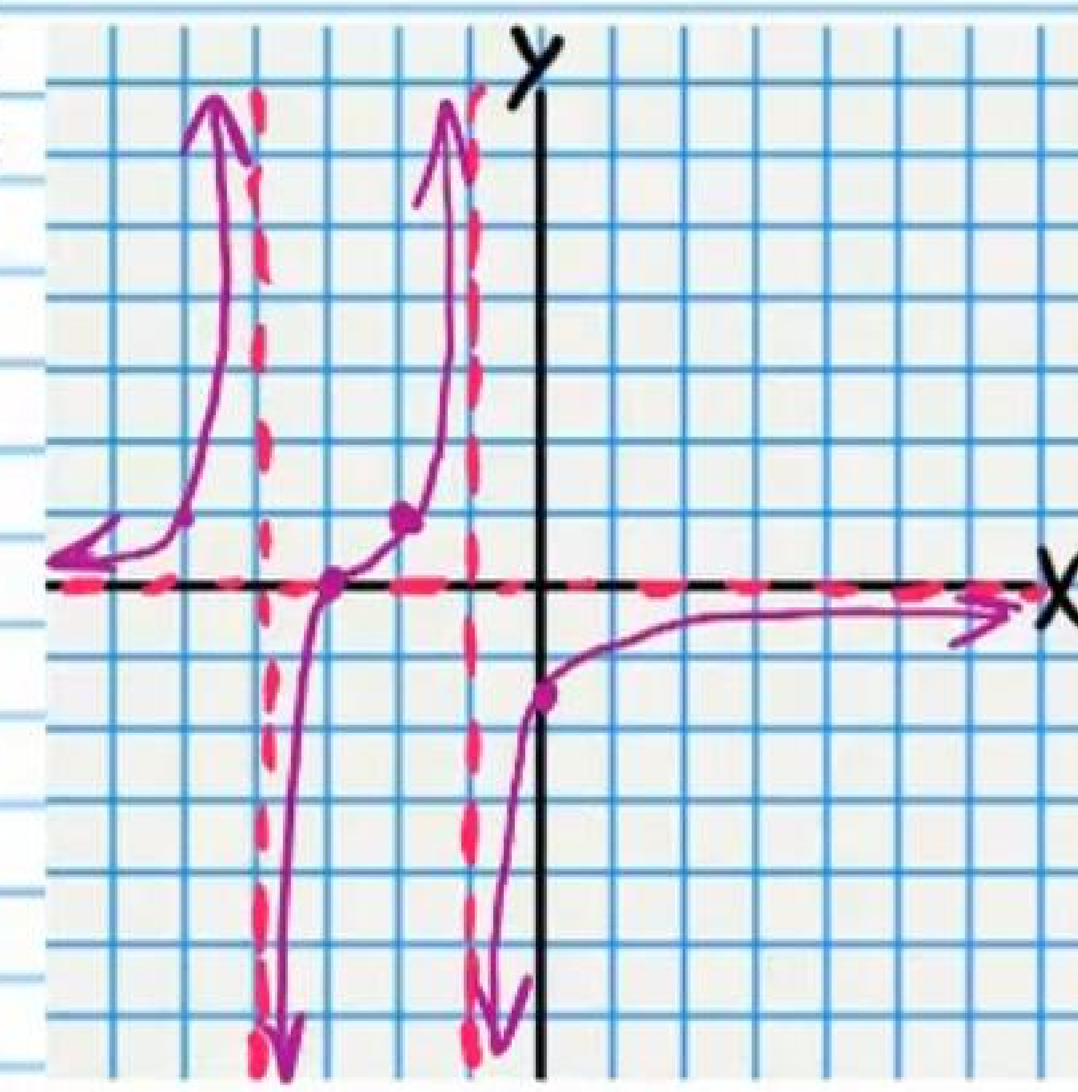
Sign Chart

$+$ | -4 | $-$ | -3 | $+$ | -1 | $-$

Y-Intercept

$f(0) = -\frac{2(0)+6}{0^2+5(0)+4} = -\frac{3}{2}$

$\left(0, -\frac{3}{2}\right)$



X-Intercept

$f(x)=0$

$-\frac{2x+6}{x^2+5x+4} = 0$

$2x+6=0$

$x=-3 \quad (-3,0)$

⑨ $f(x) = \dfrac{x+1}{x^2-4x-5}$

$f(x) = \dfrac{x+1}{(x-5)(x+1)}$

$g(x) = \dfrac{1}{x-5}$

Hole

$x = -1$

$g(-1) = \dfrac{1}{-1-5} = -\dfrac{1}{6}$

$\left(-1, -\frac{1}{6}\right)$

Vertical Asymptote Equations

$x = 5$

Horizontal Asymptote Equations

$y = 0$

Sign Chart

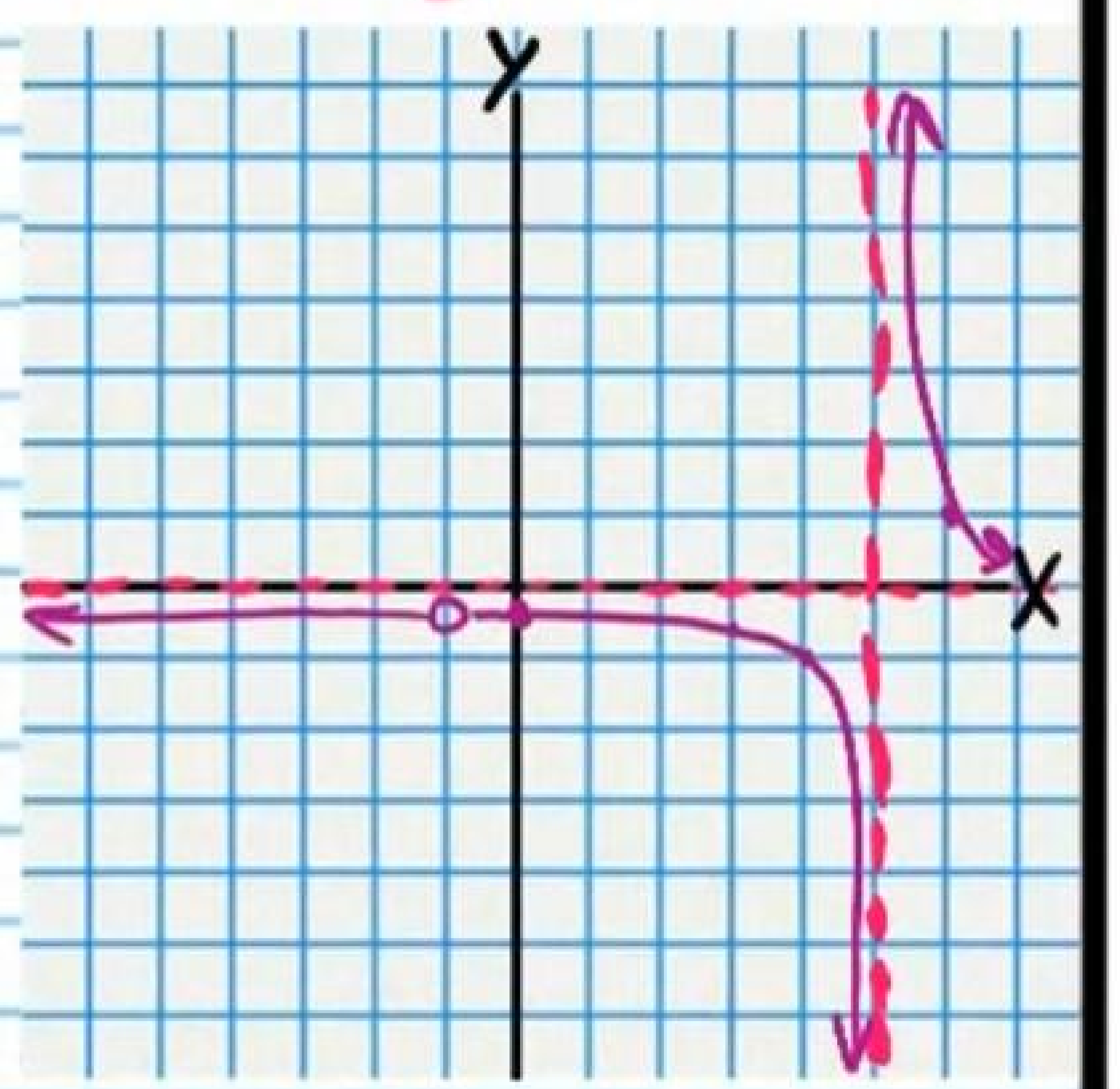
$-$ | $+$

5

Y-Intercept

$f(0) = \dfrac{0+1}{0^2-4(0)-5} = -\dfrac{1}{5}$

$\left(0, -\frac{1}{5}\right)$



X-Intercept

None

⑩ $f(x) = \dfrac{3x-15}{x^2-3x-10}$

$f(x) = \dfrac{3(x-5)}{(x+2)(x-5)}$

$g(x) = \dfrac{3}{x+2}$

Hole

$x = 5$

$g(5) = \dfrac{3}{5+2} = \dfrac{3}{7}$

$\left(5, \frac{3}{7}\right)$

Vertical Asymptote Equations

$x = -2$

Horizontal Asymptote Equations

$y = 0$

Sign Chart

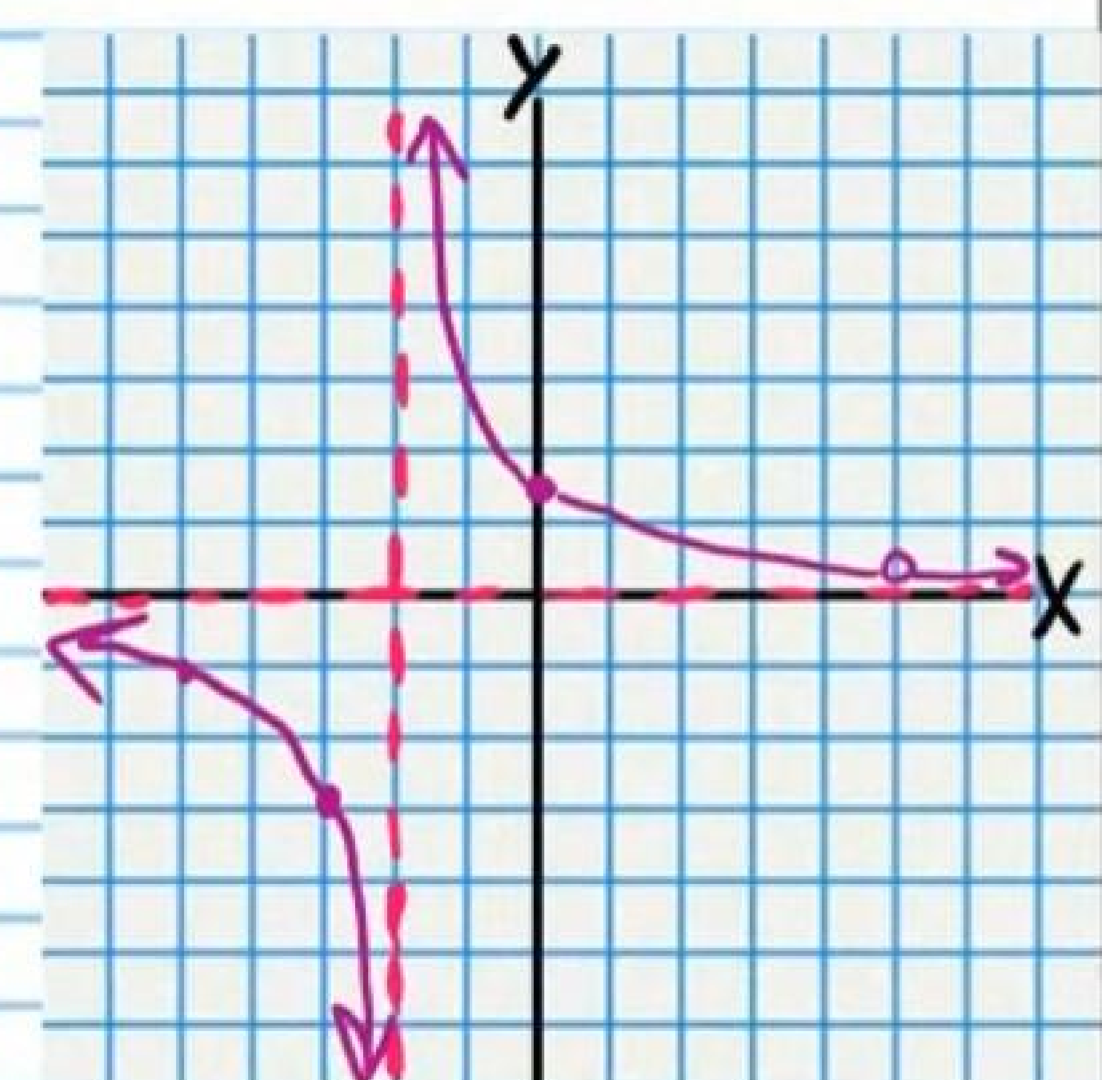
$-$ | $+$

-2

Y-Intercept

$f(0) = \dfrac{3(0)-15}{0^2-3(0)-10} = \dfrac{3}{2}$

$\left(0, \frac{3}{2}\right)$



X-Intercept

None

(11) $f(x) = \frac{x+3}{x^2+4x+3}$

$f(x) = \frac{x+3}{(x+3)(x+1)}$

$g(x) = \frac{1}{x+1}$

Hole

$x = -3$

$g(-3) = \frac{1}{-3+1} = -\frac{1}{2}$

$(-3, -\frac{1}{2})$

Vertical Asymptote Equations

$x = -1$

Horizontal Asymptote Equations

$y = 0$

Sign Chart

$-$ | $+$

-1

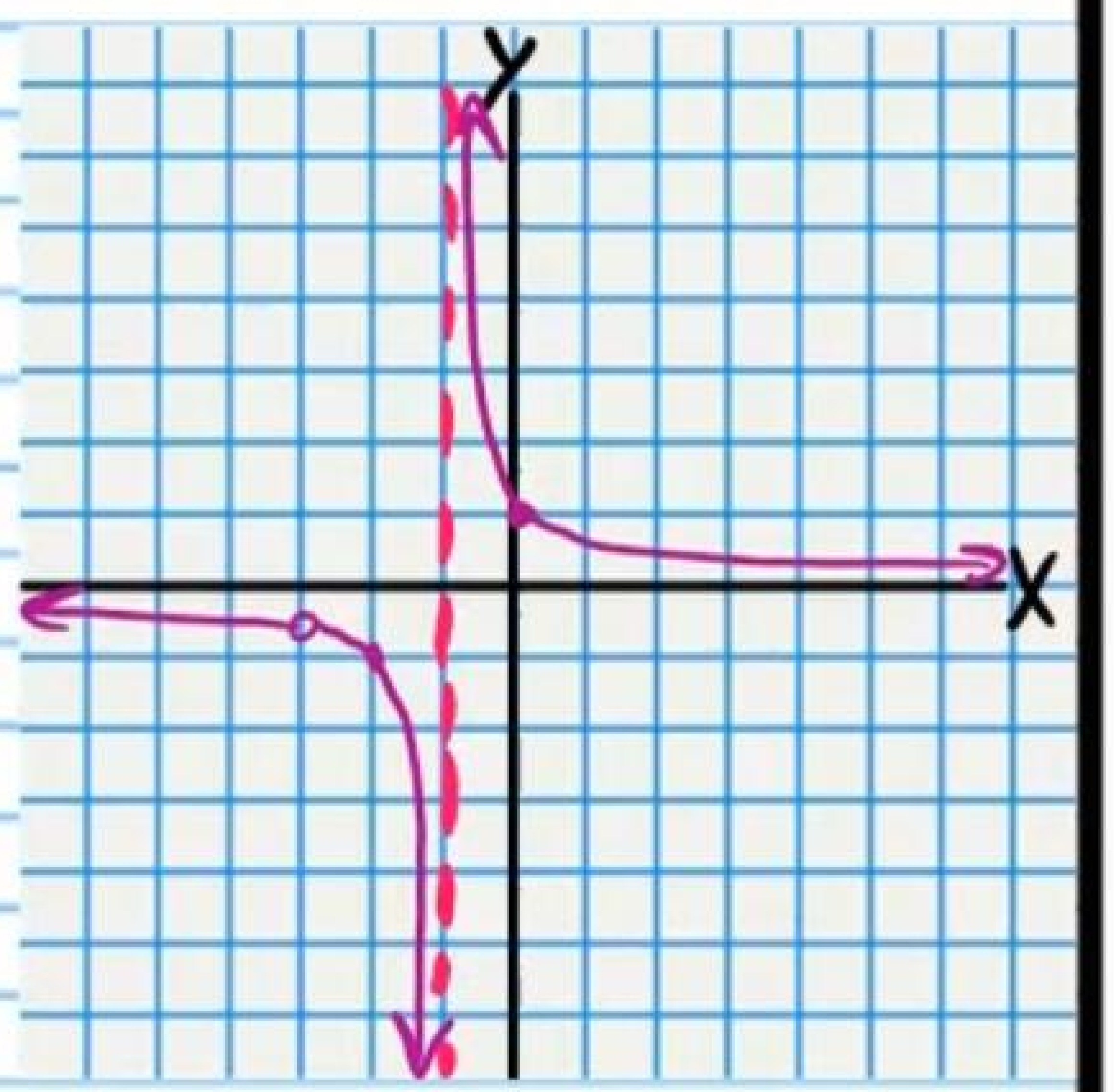
Y-Intercept

$f(0) = \frac{0+3}{0^2+4(0)+3} = 1$

$(0, 1)$

X-Intercept

None



(12) $f(x) = -\frac{2x-4}{x^2-6x+8}$

$f(x) = -\frac{2(x-2)}{(x-4)(x-2)}$

$g(x) = -\frac{2}{x-4}$

Hole

$x = 2$

$g(2) = -\frac{2}{2-4} = 1$

$(2, 1)$

Vertical Asymptote Equations

$x = 4$

Horizontal Asymptote Equations

$y = 0$

Sign Chart

$+$ | $-$

4

Y-Intercept

$f(0) = -\frac{2(0)-4}{0^2-6(0)+8} = \frac{1}{2}$

$(0, \frac{1}{2})$

X-Intercept

None